Extremal Multiplicity Profiles for Integer Horizontal Chords

Classification and Polynomial Realization

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Abstract

Let $F : \mathbb{R} \to \mathbb{R}$ be continuous. For $h > 0$, write

$$\Delta_h F(x) = F(x+h) - F(x), \qquad \nu_h(F) = \#\{x \in \mathbb{R} : \Delta_h F(x) = 0\},$$

where zeros are counted as distinct real points. For an integer $n \geq 1$, put

$$m_n(F) = \nu_n(F), \quad M(F) = \sum_{n \geq 1} m_n(F).$$

A 2025 problem from the final round of the All-Russian Mathematical Olympiad asks, for a prescribed chord of length 2025, for the least possible finite value of $M(F)$; the official answer is 4049.

We develop the equality theory for an arbitrary target N. First, $m_N(F) > 0$ and $M(F) < \infty$ imply

$$M(F) \geq 2N - 1.$$

We then completely classify the multiplicity vectors at equality. They are exactly the two families

$$V_{N,r} = (4^r, 2^{\,N-2r-1}, 0^r, 1), \qquad 0 \leq r < N/2,$$

and

$$W_{N,r} = (4^r, 3, 2^{\,N-2r-3}, 1, 0^r, 1), \qquad r \geq 0, \quad 2r+3 \leq N.$$

Here a^k denotes k consecutive entries equal to a. Consequently the number of extremal vectors is

$$2\left\lceil \frac{N}{2} \right\rceil - 1,$$

equal to N for odd N and $N-1$ for even N. The transverse extremal vectors are precisely the $V_{N,r}$.

The realization problem is also solved completely. We construct a two-parameter C^2 model by prescribing three inverse branches and their horizontal separations, obtaining the exact real-length formula

$$\nu_h = 2\mathbf{1}_{\{h<c\}} + 2\mathbf{1}_{\{h<e\}} + \mathbf{1}_{\{h=c\}} + \mathbf{1}_{\{h=e\}} + \mathbf{1}_{\{h=N\}}, \qquad c + e = N.$$

A positivity-preserving polynomialization, based on a fixed critical-point factor and two exact moment constraints, then shows that every $V_{N,r}$ and every $W_{N,r}$ is realizable by a polynomial with rational coefficients. Thus the extremal support, the nontransverse multiplicity allocation, and polynomial existence are all determined for every target.

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1 Introduction

A horizontal chord of a real-valued function is a horizontal line segment whose endpoints lie on the graph. Equivalently, a chord of length $h > 0$ is a solution of

$$F(x + h) = F(x).$$

The classical horizontal-chord theory asks which *lengths* occur. The universal chord theorem of Lévy and the later theorem of Hopf are fundamental results in this direction; modern accounts include Oxtoby [3], Burns–Davidovich–Davis [4], and Davis–Troubetzkoy [5]. Recent work of Ciudin–Ionascu [6] continues the study of the topology and isolated points of chord sets.

The present problem is different. We restrict attention to *integer* lengths and ask not only whether a length occurs, but how many distinct chords of that length occur. Thus the basic object is the multiplicity sequence

$$(m_1(F), m_2(F), \ldots).$$

This refinement was prompted directly by Problem 11.8 of the 2025 final round of the All-Russian Mathematical Olympiad, proposed by A. S. Kuznetsov. For a continuous $F : \mathbb{R} \to \mathbb{R}$ with finitely many integer horizontal chords and with at least one chord of length 2025, the problem asks for the smallest possible total number of chords. The official solution proves that the answer is 4049 [7].

The first part of the olympiad argument extends naturally from 2025 to an arbitrary target N, giving the sharp minimum $2N - 1$. The purpose of this paper is to understand the much more rigid equality case. Three questions drive the analysis:

(i) Which integer lengths may be absent when $M(F) = 2N - 1$?

(ii) Once the support is known, which multiplicities can actually occur without a transversality assumption?

(iii) Can every admissible extremal vector be realized by a polynomial, uniformly in N?

All three questions admit complete answers. The missing lengths below N form a terminal interval. Global compatibility among the difference functions then reduces the apparent local choices $(1,3),(2,2),(3,1)$ to a single possible exceptional pair, yielding the two families V and W. Finally, an inverse-branch construction realizes all of them at once, and a positive polynomial weight with two exact moments converts the construction into one with rational polynomial coefficients.

1.1 Relation to the existing literature

The distinction between chord *support* and chord *multiplicity* is important. Hopf-type results characterize, or constrain, the set

$$\mathcal{H}(F) = \{h > 0 : \nu_h(F) > 0\},$$

whereas the present work studies the cardinalities of the fibers $\{x : F(x+h) = F(x)\}$ at integer h, under a finite-total constraint. The papers cited above are therefore closely related background, but their principal object is different.

The closest direct predecessor we located is the 2025 olympiad problem and its official solution, which proves the sharp total for the single target 2025 and contains the complementary-length mechanism used below. A literature search conducted through September 2026 located no published paper or arXiv preprint whose main object is the finite integer multiplicity sequence $n \mapsto m_n(F)$, nor one containing the extremal V/W classification or the all-parameter polynomial realization theorem proved here. This is a report of the search outcome, not a claim of absolute priority over all possible literature.

2 Definitions and elementary identities

Definition 2.1. For a continuous function $F : \mathbb{R} \to \mathbb{R}$ and $h > 0$, define

$$\Delta_h F(x) := F(x+h) - F(x), \qquad \nu_h(F) := \#\{x \in \mathbb{R} : \Delta_h F(x) = 0\}.$$

If the zero set is infinite, we write $\nu_h(F) = \infty$. For $n \in \mathbb{N}$,

$$m_n(F) := \nu_n(F), \qquad M(F) := \sum_{n \geq 1} m_n(F) \in \mathbb{N} \cup \{\infty\}.$$

Definition 2.2. Fix $N \in \mathbb{N}$. We call F *N-extremal* if

$$m_N(F) > 0, \qquad M(F) = 2N - 1.$$

If $F \in C^1(\mathbb{R})$, we call it *transverse below N* if every zero of $\Delta_n F$ is simple for $1 \leq n < N$.

We repeatedly use the cocycle identities

$$\Delta_{a+b} F(x) = \Delta_a F(x) + \Delta_b F(x+a), \tag{1}$$

$$\Delta_n F(x) = \sum_{j=0}^{n-1} \Delta_1 F(x+j). \tag{2}$$

Lemma 2.3 (Common tail orientation)**.** *If $M(F) < \infty$, then after replacing F by $-F$, if necessary,*

$$\Delta_1 F(x) > 0$$

for all sufficiently large $|x|$. Consequently, for every fixed $n \geq 1$, $\Delta_n F(x) > 0$ on both sufficiently remote tails.

Proof. Since $m_1(F) < \infty$, the continuous function $g = \Delta_1 F$ is nonzero and has a constant sign on each sufficiently remote tail. If those two signs were opposite, then (2) would give opposite tail signs for every $\Delta_n F$. The intermediate value theorem would then imply $m_n(F) > 0$ for every n, contradicting $M(F) < \infty$. Thus the two tail signs agree, and a global sign change of F makes them positive. Equation (2) gives the assertion for each fixed n. □

Two immediate consequences will be used without further comment:

$$m_d(F) = 0 \Longrightarrow \Delta_d F > 0, \qquad m_d(F) = 1 \Longrightarrow \Delta_d F \geq 0, \tag{2.1}$$

after the orientation of Lemma 2.3.

3 The sharp total and equality constraints

Theorem 3.1 (Complementary-length inequality)**.** *Suppose $M(F) < \infty$, $m_N(F) > 0$, and $a, b \in \mathbb{N}$ satisfy $a + b = N$. Then*

$$m_a(F) + m_b(F) \geq 4.$$

Proof. Translate the variable and subtract a constant so that one target chord is

$$F(0) = F(N) = 0.$$

Orient the tails by Lemma 2.3. Suppose, to the contrary, that $m_a + m_b \leq 3$. Interchanging a, b, we may assume $m_a \leq 1$. By (2) and the positive tails, a negative value of $\Delta_a F$ would force at least two zeros; hence

$$\Delta_a F \geq 0. \tag{3.1}$$

If $a = b$, then

$$0 = \Delta_N F(0) = \Delta_a F(0) + \Delta_a F(a),$$

so both terms vanish, contradicting $m_a \leq 1$. Hence $a \neq b$.

Let $q = N/\gcd(a, N)$. For $1 \leq j \leq q - 1$, let r_j be the representative of $ja \bmod N$ in $\{1, \ldots, N - 1\}$. Choose

$$r_0 = r_q = \begin{cases} 0, & \Delta_a F(0) > 0, \\ N, & \Delta_a F(0) = 0. \end{cases}$$

Every step $r_j \to r_{j+1}$ either increases by a or decreases by $b = N - a$. Put

$$d_j := F(r_{j+1}) - F(r_j).$$

An increasing step gives $d_j = \Delta_a F(r_j) \geq 0$. At least one such step is strictly positive: it is the first step when $r_0 = 0$, and the last step when $r_0 = N$, because in the latter case the unique possible zero of $\Delta_a F$ is already at 0.

Since

$$\sum_{j=0}^{q-1} d_j = F(r_q) - F(r_0) = 0,$$

some decreasing step has $d_j < 0$. If its landing point is t, then

$$\Delta_b F(t) > 0, \qquad 0 < t < a. \tag{3.2}$$

On the other hand,

$$\Delta_b F(0) = -\Delta_a F(b) \le 0, \qquad \Delta_b F(a) = -\Delta_a F(0) \le 0. \tag{3.3}$$

If both values in (3.3) are negative, then the positive tails, the two negative endpoint values, and (3.2) force four zeros of $\Delta_b F$. If exactly one endpoint value is zero, then $\Delta_b F$ has at least three zeros and $\Delta_a F$ has at least one. Both endpoint values cannot vanish, for that would give two distinct zeros of $\Delta_a F$. In all cases $m_a + m_b \ge 4$, a contradiction. □

Theorem 3.2 (Sharp total)**.** *If $m_N(F) > 0$ and $M(F) < \infty$, then*

$$M(F) \ge 2N - 1.$$

The bound is sharp.

Proof. Pair the lengths $1, \dots, N-1$ by $a \leftrightarrow N-a$ and apply Theorem 3.1. For odd N, the $(N-1)/2$ pairs contribute at least 4 each, and the target contributes at least 1. For even N, the unequal pairs contribute 4 each, while $a = N/2$ contributes at least 2. In both cases the total is at least $2N - 1$.

For sharpness, let

$$P_N(x) = x^3 - \frac{N^2}{4}x.$$

Then

$$\frac{\Delta_n P_N(x)}{n} = 3\left(x + \frac{n}{2}\right)^2 + \frac{n^2 - N^2}{4}.$$

Thus $m_n(P_N) = 2$ for $1 \le n < N$, $m_N(P_N) = 1$, and $m_n(P_N) = 0$ for $n > N$. Hence $M(P_N) = 2N - 1$. □

Therefore every N-extremal function satisfies

$$m_N = 1, \qquad m_n = 0 \ \ (n > N), \qquad m_a + m_{N-a} = 4 \ \ (1 \le a < N). \tag{E}$$

4 Missing lengths and terminal support

The missing lengths have an additive structure even before extremality.

Theorem 4.1 (Missing-length semigroup)**.** *If $M(F) < \infty$, then*

$$\Gamma(F) := \{0\} \cup \{n \in \mathbb{N} : m_n(F) = 0\}$$

is a numerical semigroup.

Proof. Only additive closure requires proof. If a, b are missing, then $\Delta_a F$ and $\Delta_b F$ never vanish and therefore each has a constant sign. Their signs must agree, since

$$\Delta_{ab} F(x) = \sum_{j=0}^{b-1} \Delta_a F(x + ja) = \sum_{j=0}^{a-1} \Delta_b F(x + jb)$$

cannot simultaneously have opposite signs. Hence

$$\Delta_{a+b} F(x) = \Delta_a F(x) + \Delta_b F(x + a)$$

never vanishes, so $a + b$ is missing. Since $M(F) < \infty$, only finitely many positive integers occur, and $\Gamma(F)$ is cofinite. □

For the equality case, much more is true.

Lemma 4.2 (Strict block propagation)**.** *Assume $F(0) = F(N)$, $M(F) < \infty$, and let $1 \le d < N$ be missing. Write*

$$N = qd + s, \qquad 0 \le s < d.$$

Then $s \neq 0$. If $q \ge 2$, then

$$m_1(F) \ge 2(q + 1).$$

Proof. Orient the tails positively and set $g = \Delta_1 F$, $g_j = g(j)$. Since $\Delta_d F > 0$, if $s = 0$ then $\Delta_N F(0)$ is a sum of q positive d-blocks, impossible. Thus $1 \le s < d$.

For $t = 0, \dots, q$, let

$$R_t = \{td, \dots, td + s - 1\}.$$

The complement of R_t in $\{0, \dots, N - 1\}$ is a disjoint union of q consecutive d-blocks, each of positive sum. Since $\sum_{j=0}^{N-1} g_j = 0$, every R_t has negative sum. Between R_t and R_{t+1} lies

$$G_t = \{td + s, \dots, (t + 1)d - 1\}, \qquad 0 \le t < q.$$

Because $R_t \cup G_t$ is a positive d-block, G_t has positive sum.

Choose one negative sample from each R_t and one positive sample from each G_t. These chosen signs alternate, beginning and ending negative. Together with the positive tails, the intermediate value theorem forces $2(q + 1)$ distinct zeros of g. □

Lemma 4.3 (Weak gap filling)**.** *Assume $F(0) = F(N)$, $M(F) < \infty$, and the positive tail orientation. Let*

$$2 \le k < s < N/2.$$

Suppose

$$m_1 = 4, \qquad m_{N-j} = 0 \ \ (1 \le j < k), \qquad \Delta_{N-s} F \ge 0.$$

Then

$$m_k \ge 4.$$

Proof. Put $g = \Delta_1 F$, $g_j = g(j)$, and define

$$P_u = \sum_{j=0}^{u-1} g_j, \qquad Q_u = \sum_{j=N-u}^{N-1} g_j.$$

The target equation gives $\sum_{j=0}^{N-1} g_j = 0$. If $\Delta_{N-u}F \geq 0$, evaluation at 0 and at u gives

$$P_u \leq 0, \qquad Q_u \leq 0, \tag{4.1}$$

with strict inequalities when $N-u$ is missing. Hence

$$P_j, Q_j < 0 \quad (1 \leq j < k), \qquad P_s, Q_s \leq 0. \tag{4.2}$$

In particular $g_0, g_{N-1} < 0$.

We claim $P_k < 0$. Suppose $P_k \geq 0$. Since $P_{k-1} < 0$, we have $g_{k-1} > 0$. Also

$$\sum_{j=s}^{N-1} g_j = -P_s \geq 0,$$

and $g_{N-1} < 0$, so some $g_t > 0$ with $t \geq s$. The two positive samples g_{k-1} and g_t, together with the negative endpoint samples and the positive tails, already account for the four zeros allowed by $m_1 = 4$. Therefore every sampled value between those two positive samples must itself be strictly positive; otherwise an additional zero or two additional sign changes would occur. Hence $g_k, \ldots, g_{s-1} > 0$, and

$$P_s = P_k + \sum_{j=k}^{s-1} g_j > 0,$$

contradicting (4.1). Thus $P_k < 0$. Reversing the sample order gives $Q_k < 0$.

Now set

$$T_j = \sum_{\ell=0}^{k-1} g_{j+\ell} = \Delta_k F(j), \qquad 0 \leq j \leq N-k.$$

Then

$$T_0 = P_k < 0, \qquad T_{N-k} = Q_k < 0.$$

Counting the number of times each sample occurs in the sliding sums,

$$\sum_{j=0}^{N-k} T_j = -\sum_{u=1}^{k-1} (P_u + Q_u) > 0.$$

Thus some interior T_j is positive. Since $\Delta_k F$ has positive tails, the sampled sign pattern negative–positive–negative forces four distinct zeros. Hence $m_k \geq 4$. □

Theorem 4.4 (Terminal support)**.** *Let F be N-extremal. Then there is a unique integer r with $0 \leq r < N/2$ such that*

$$\{1 \leq d < N : m_d = 0\} = \{N-r, N-r+1, \ldots, N-1\}. \tag{4.3}$$

The set on the right is empty when $r = 0$.

Proof. Let $A = \{1 \leq d < N : m_d = 0\}$. If $d < N/2$ belongs to A, Lemma 4.2 gives $m_1 \geq 6$, contradicting (E). The value $d = N/2$ is also impossible, because a missing half-length would make $\Delta_N F$ a sum of two positive half-length differences. Therefore $A \subset (N/2, N)$.

If $A = \varnothing$, take $r = 0$. Otherwise choose $d \in A$. Lemma 4.2, now with $q = 1$, gives $m_1 \geq 4$; hence (E) forces

$$m_1 = 4, \qquad m_{N-1} = 0.$$

Let $\min A = N - r$. Then $1 \leq r < N/2$. We prove inductively that $N - k \in A$ for $1 \leq k \leq r$. The case $k = 1$ is known. If $2 \leq k < r$ and $N - j \in A$ for every $j < k$, apply Lemma 4.3 with $s = r$. Since $N - r$ is missing, $\Delta_{N-r}F > 0$, so $m_k \geq 4$. The equality $m_k + m_{N-k} = 4$ then gives $m_{N-k} = 0$. Finally $N - r \in A$ by definition. Minimality of $N - r$ excludes every other missing length below it. □

5 Single zeros and the complete extremal classification

Terminal support still leaves a formal possibility: in a central complementary pair, equality in (E) allows

$$(1,3), \quad (2,2), \quad (3,1).$$

The next lemma shows that a single zero is highly constrained.

Lemma 5.1 (Weak block propagation for a single zero)**.** *Assume $F(0) = F(N)$, $M(F) < \infty$, and $m_d = 1$ for some $d < N$.*

(a) If $d \mid N$, a contradiction follows.

(b) If $q = \lfloor N/d \rfloor \geq 2$, then

$$m_1 \geq 2(q+1).$$

(c) If $d > N/2$ and $m_1 \leq 3$, then

$$d = N - 1, \qquad m_1 = 3.$$

Proof. Orient the tails positively. Since $m_d = 1$, $\Delta_d F \geq 0$.

If $N = qd$, the target identity is a sum of $q \geq 2$ nonnegative d-differences. For the sum to vanish, all q terms must vanish at distinct starting points, contradicting $m_d = 1$. This proves (a).

For (b), write $N = qd + s$, $1 \leq s < d$, and repeat the block notation from Lemma 4.2. The complement of each remainder block R_t is a union of q nonnegative d-blocks. At most one of those blocks can have zero sum, while $q \geq 2$, so the complement has strictly positive sum and R_t has strictly negative sum. Because $R_t \cup G_t$ is a nonnegative d-block, G_t has strictly positive sum. The same alternating sign argument yields $m_1 \geq 2(q+1)$.

For (c), write $N = d + s$, $1 \leq s < d$. The two remainder blocks

$$R_0 = \{0, \ldots, s-1\}, \qquad R_1 = \{d, \ldots, N-1\}$$

have nonpositive sums, and at least one has strictly negative sum; otherwise $\Delta_d F$ would vanish at two distinct starting points. The middle block

$$G = \{s, \ldots, d-1\}$$

therefore has positive sum. If both remainder blocks contain a negative sample, then negative–positive–negative, together with the positive tails, forces $m_1 \geq 4$. Thus, under $m_1 \leq 3$, one remainder block contains no negative sample. Since its sum is nonpositive, all its s samples are zero. The positive middle block and the negative other remainder block force two further zeros of $\Delta_1 F$, distinct from those s sampled zeros. Hence $m_1 \geq s + 2$. Therefore $s = 1$ and $m_1 = 3$, i.e. $d = N - 1$. □

Corollary 5.2. *For an N-extremal function, a coordinate $m_d = 1$ with $d < N$ can occur only for*

$$N/2 < d < N.$$

Proof. If $d \leq N/2$, either $d \mid N$ or $\lfloor N/d \rfloor \geq 2$. Lemma 5.1 then contradicts $m_1 \leq 4$, which follows from (E). □

We can now state the main classification theorem.

Theorem 5.3 (Complete extremal multiplicity classification)**.** *Let $F : \mathbb{R} \to \mathbb{R}$ be continuous and N-extremal. Then its vector*

$$(m_1(F), \dots, m_N(F))$$

is exactly one of the following two forms:

$$V_{N,r} = (4^r, 2^{N-2r-1}, 0^r, 1), \qquad 0 \le r < N/2, \tag{5.1}$$

or

$$W_{N,r} = (4^r, 3, 2^{N-2r-3}, 1, 0^r, 1), \qquad r \ge 0, \quad 2r+3 \le N. \tag{5.2}$$

Proof. Let r be given by Theorem 4.4. The missing terminal block forces, by (E),

$$m_n = 4 \quad (1 \le n \le r), \qquad m_n = 0 \quad (N-r \le n < N).$$

Every central coordinate $r < n < N-r$ is positive. In a central complementary pair, the two multiplicities are positive integers summing to 4. By Corollary 5.2, the only possibilities, written in short-length/long-length order, are

$$(2,2) \quad \text{or} \quad (3,1). \tag{5.3}$$

First suppose $r = 0$. If no single-zero coordinate occurs, every pair is $(2,2)$, giving $V_{N,0}$. If a single-zero coordinate occurs, Lemma 5.1(c) forces it to be $N-1$ and forces $m_1 = 3$. Hence the unique exceptional pair is $(3,1)$, giving $W_{N,0}$.

Now suppose $r \ge 1$, so $m_1 = 4$. Assume a central long length $d = N - s$ has $m_d = 1$. Since d is central,

$$r < s < N/2.$$

If $s \ge r+2$, choose $k = r+1 < s$. The terminal support gives $m_{N-j} = 0$ for every $1 \le j < k$, while $m_{N-s} = 1$ gives $\Delta_{N-s}F \ge 0$. Lemma 4.3 yields $m_{r+1} \ge 4$, and then (E) forces

$$m_{N-r-1} = 0,$$

contradicting the fact that the missing block starts at $N-r$. Therefore $s = r+1$. Thus at most one exceptional $(3,1)$ pair can occur, and it must be exactly

$$(r+1, N-r-1).$$

The two lengths are distinct precisely when $2r+3 \le N$. This gives $W_{N,r}$; otherwise all central pairs are $(2,2)$, giving $V_{N,r}$. □

Corollary 5.4 (Number of extremal vectors)**.** *The exact number of N-extremal multiplicity vectors is*

$$\left\lceil \frac{N}{2} \right\rceil + \left\lfloor \frac{N-1}{2} \right\rfloor = 2\left\lceil \frac{N}{2} \right\rceil - 1.$$

Thus it equals N when N is odd and $N-1$ when N is even.

The theorem above proves necessity. The rest of the paper proves that every listed vector is realizable, even polynomially.

6 An inverse-branch model with exact real chord counts

We first construct a C^2 function whose *entire real-length* chord count is explicit.

Fix real numbers

$$c, e \geq \frac{1}{2}, \qquad c + e = N.$$

Set

$$\delta = \frac{1}{8}, \qquad \alpha = \frac{1}{16}, \qquad \gamma = \frac{\delta}{\sqrt{3}}, \qquad v = c - \delta, \qquad w = c + \delta.$$

For $b \geq 1/2$, define

$$I_b(t) = \alpha\sqrt{t} + \gamma(1 - \sqrt{1-t}) + (b - \delta - \alpha - \gamma)t, \qquad 0 \leq t \leq 1,$$

and

$$\theta(t) = 1 - \sqrt{1 - t^{1/3}}.$$

The coefficient $b - \delta - \alpha - \gamma$ is positive, hence I_b is strictly increasing, with

$$I_b(0) = 0, \qquad I_b(1) = b - \delta.$$

Define $H_{N,c} : \mathbb{R} \to \mathbb{R}$ by

$$H_{N,c}(x) = \begin{cases} -(x/\alpha)^2, & x \leq 0, \\ -I_c^{-1}(x), & 0 \leq x \leq v, \\ \operatorname{sgn}(x-c)\left(\dfrac{2|x-c|}{\delta} - \left(\dfrac{x-c}{\delta}\right)^2\right)^3, & v \leq x \leq w, \\ I_e^{-1}(N-x), & w \leq x \leq N, \\ ((x-N)/\alpha)^2, & x \geq N. \end{cases} \tag{6.1}$$

Proposition 6.1 (Shape and regularity)**.** *The function $H_{N,c}$ is C^2. Its strict turning points are*

$$0, \quad v, \quad w, \quad N,$$

and c is a horizontal inflection point on the increasing middle branch. Moreover

$$H(0) = H(c) = H(N) = 0, \qquad H(v) = -1, \qquad H(w) = 1,$$

and

$$H''(0) = -\frac{2}{\alpha^2}, \quad H''(v) = \frac{6}{\delta^2}, \quad H''(w) = -\frac{6}{\delta^2}, \quad H''(N) = \frac{2}{\alpha^2}.$$

Near c,

$$H(c+z) = \frac{8}{\delta^3}z^3 + O(|z|^4).$$

Proof. Monotonicity on the five branches is immediate from the definitions. Near $t = 0$,

$$I_b(t) = \alpha\sqrt{t} + O(t),$$

so inversion gives the same quadratic germ as the outer parabola. Near $t = 1$,

$$I_b(t) = b - \delta - \gamma\sqrt{1-t} + O(1-t),$$

hence inversion gives a quadratic germ with second derivative $2/\gamma^2 = 6/\delta^2$, matching the middle branch at v and w. Finally the middle formula gives

$$H(c+z) = 8z^3/\delta^3 + O(|z|^4),$$

so the first two derivatives match at c as well. □

Theorem 6.2 (Exact real-length chord count)**.** *For every real $h > 0$,*

$$\nu_h(H_{N,c}) = 2\mathbf{1}_{\{h<c\}} + 2\mathbf{1}_{\{h<e\}} + \mathbf{1}_{\{h=c\}} + \mathbf{1}_{\{h=e\}} + \mathbf{1}_{\{h=N\}}. \tag{6.2}$$

The two indicators at c and e are counted separately when $c = e$. Every nonzero-height chord gives a simple zero of the corresponding difference function. The zero-height chords are exactly

$$(0,c), \qquad (c,N), \qquad (0,N),$$

and they are strict quadratic tangencies of the relevant differences.

Proof. Consider first a negative level $-t$, $0 < t < 1$. Its three intersection points, from left to right, are

$$x_1 = -\alpha\sqrt{t}, \qquad x_2 = I_c(t), \qquad x_3 = c - \delta\theta(t).$$

Their three pairwise distances are

$$\begin{aligned} D_1(t) &= I_c(t) + \alpha\sqrt{t}, \\ D_2(t) &= c - \delta\theta(t) - I_c(t), \\ D_3(t) &= c - \delta\theta(t) + \alpha\sqrt{t}. \end{aligned}$$

Put

$$d_c = c - \delta + \alpha.$$

Then D_1 is strictly increasing from 0 to d_c, while D_2 is strictly decreasing from c to 0. For D_3,

$$\theta'(t) = \frac{1}{6t^{2/3}\sqrt{1-t^{1/3}}},$$

and therefore

$$2\sqrt{t}\,\delta\theta'(t) = \frac{\delta}{3\sqrt{t^{1/3}(1-t^{1/3})}} \geq \frac{2\delta}{3} > \alpha.$$

Hence D_3 is strictly decreasing from c to d_c.

It follows that each $0 < h < c$ occurs exactly twice as a negative-level distance. At $h = d_c$, the endpoint values of D_1 and D_3 represent the same chord, while D_2 contributes one further chord, so the count remains two. At $h = c$, the only negative-side limit is the zero-level chord $(0,c)$. No $h > c$ occurs on a negative level.

The positive-level picture is the reflection of the same calculation with c replaced by e; it contributes two chords for $0 < h < e$, one zero-level chord at $h = e$, and none for $h > e$. The only remaining zero-level chord is $(0,N)$. This proves (6.2).

For a nonzero level, a horizontal separation varying strictly with the level implies that the two endpoint derivatives are unequal, so the corresponding zero of $\Delta_h H$ is simple. At the merged level $t = 1$, one endpoint is a turning point and the other has nonzero derivative, so simplicity still holds. For the zero-level chords, both first derivatives vanish, while

$$(\Delta_c H)''(0) = \frac{2}{\alpha^2}, \qquad (\Delta_e H)''(c) = \frac{2}{\alpha^2}, \qquad (\Delta_N H)''(0) = \frac{4}{\alpha^2}.$$

Thus all three are strict quadratic tangencies. □

Corollary 6.3 (All extremal profiles in C^2)**.** *For*

$$c = N - r - \frac{1}{2}, \qquad e = r + \frac{1}{2},$$

the function $H_{N,c}$ realizes $V_{N,r}$. For

$$c = N - r - 1, \qquad e = r + 1,$$

with $2r + 3 \le N$, it realizes $W_{N,r}$. In the V-case, every zero below the target is simple.

Proof. Insert the two parameter choices into (6.2). In the V-case, neither c nor e is an integer. In the W-case, the integer lengths e and c each acquire one quadratic tangency, which changes the adjacent pair from $(2, 2)$ to $(3, 1)$. □

7 Polynomialization by a positive derivative weight

A naive uniform polynomial approximation of $H_{N,c}$ is insufficient: it may split a quadratic tangency into two zeros, remove it, or create new long chords far from the approximation interval. We instead preserve the critical-point geometry exactly.

Assume from now on that $N \in \mathbb{N}$ and that $c, e = N - c \ge 1/2$ are rational. Let

$$J = \left[-\frac{1}{4}, N + \frac{1}{4}\right]$$

and define

$$\Omega(x) = x(x - v)(x - c)^2(x - w)(x - N). \tag{7.1}$$

By Proposition 6.1,

$$U(x) := \frac{H'_{N,c}(x)}{\Omega(x)}$$

extends continuously across the five zeros of Ω, and

$$U(x) > 0 \qquad (x \in J).$$

Moreover,

$$\int_0^c \Omega(x)U(x)\,dx = 0, \qquad \int_c^N \Omega(x)U(x)\,dx = 0. \tag{7.2}$$

Lemma 7.1 (Positive rational polynomial with two exact moments)**.** *There exist polynomials $U_j \in \mathbb{Q}[x]$ such that*

$$U_j(x) > 0 \quad \textit{for all } x \in \mathbb{R}, \qquad U_j \longrightarrow U \quad \textit{uniformly on } J,$$

and

$$\int_0^c \Omega U_j = 0, \qquad \int_c^N \Omega U_j = 0. \tag{7.3}$$

Proof. Define the linear map

$$L(B) = \left(\int_0^c \Omega B, \int_c^N \Omega B\right).$$

The sign of Ω on the four intervals

$$(0, v), \quad (v, c), \quad (c, w), \quad (w, N)$$

is, respectively,

$$-, \quad +, \quad +, \quad -.$$

Choose four nonnegative continuous bump functions, one supported in each interval, normalized so that their L-vectors are

$$(-1, 0), \quad (1, 0), \quad (0, 1), \quad (0, -1).$$

Approximate the square root of each bump uniformly on J by a rational polynomial h_i. For sufficiently accurate approximation, the four vectors

$$z_i := L(h_i^2)$$

remain in the four corresponding quadrants, in cyclic order, with every adjacent angular gap strictly less than π. Hence the positive cones generated by adjacent pairs cover $\mathbb{R}^2$. Moreover there is a constant C such that any $b \in \mathbb{R}^2$ can be written as

$$b = \sum_i \lambda_i z_i, \qquad \lambda_i \geq 0, \qquad \sum_i \lambda_i \leq C\|b\|, \tag{7.4}$$

using at most two adjacent generators. If $b \in \mathbb{Q}^2$, the coefficients λ_i may be chosen rational.

Now approximate $\sqrt{U}$ uniformly on J by $p_j \in \mathbb{Q}[x]$, and choose rational $\eta_j > 0$ with $\eta_j \to 0$. Put

$$B_j = p_j^2 + \eta_j.$$

Then $B_j > 0$ on all of $\mathbb{R}$, $B_j \to U$ uniformly on J, and, by (7.2),

$$L(B_j) \to 0.$$

Since the endpoints and Ω are rational, $L(B_j) \in \mathbb{Q}^2$. Represent $-L(B_j)$ as in (7.4) and define

$$U_j = B_j + \sum_i \lambda_{j,i} h_i^2.$$

Then $U_j \in \mathbb{Q}[x]$, $U_j > 0$ on all of $\mathbb{R}$, and $L(U_j) = 0$. The bound in (7.4) gives $\lambda_{j,i} \to 0$, so $U_j \to U$ uniformly on J. □

Theorem 7.2 (Polynomial realization)**.** *Let $N \in \mathbb{N}$, let $c, e = N - c \geq 1/2$ be rational, and let $H_{N,c}$ be the function from (6.1). Then there exists a polynomial $P \in \mathbb{Q}[x]$ such that*

$$m_n(P) = m_n(H_{N,c}) \qquad \text{for every integer } n \geq 1.$$

Every simple integer-difference zero of $H_{N,c}$ remains simple for P, and every integer zero-level tangency remains a double zero.

Proof. Choose U_j from Lemma 7.1 and define

$$P_j(x) = \int_0^x \Omega(t) U_j(t)\, dt. \tag{7.5}$$

Then $P_j \in \mathbb{Q}[x]$ and the exact moments give

$$P_j(0) = P_j(c) = P_j(N) = 0. \tag{7.6}$$

Also

$$P_j'(x) = \Omega(x)U_j(x),$$

so, because $U_j > 0$ on all of $\mathbb{R}$, the complete real critical set of P_j is fixed exactly:

$$0, \quad v, \quad c, \quad w, \quad N,$$

with c a double zero of the derivative and the other four simple zeros. Thus no additional real critical point can appear outside the approximation interval.

Uniform convergence $U_j \to U$ on J implies

$$P_j \to H, \qquad P_j' \to H'$$

uniformly on J. Since

$$H\left(-\frac{1}{4}\right) = -16, \qquad H(v) = -1, \qquad H(w) = 1, \qquad H\left(N + \frac{1}{4}\right) = 16,$$

for all sufficiently large j,

$$P_j\left(-\frac{1}{4}\right) < -2 < P_j(v) < 0 < P_j(w) < 2 < P_j\left(N + \frac{1}{4}\right). \tag{7.7}$$

The fixed derivative sign pattern now implies that every horizontal level having at least two intersections with the graph of P_j has all of its intersection points in J. Consequently every horizontal chord of P_j has both endpoints in J. Since

$$|J| = N + \frac{1}{2} < N + 1,$$

there are no integer chords of length $n \geq N + 1$.

We next preserve the quadratic tangencies. The functions U_j are uniformly bounded above and away from 0 on J. Since Ω has simple zeros at $0, N$ and a double zero at c, there exist constants $A, B, \varepsilon > 0$, independent of sufficiently large j, such that for $|t| < \varepsilon$,

$$-P_j(t) \geq At^2, \qquad P_j(N + t) \geq At^2, \qquad |P_j(c + t)| \leq B|t|^3.$$

Shrink ε so that $B\varepsilon < A$. Then, for $0 < |t| < \varepsilon$,

$$\Delta_c P_j(t) > 0, \qquad \Delta_e P_j(c + t) > 0, \qquad \Delta_N P_j(t) > 0. \tag{7.8}$$

At the central points these differences vanish by (7.6). Their second derivatives there are strictly positive:

$$(\Delta_c P_j)''(0) = -P_j''(0) > 0,$$
$$(\Delta_e P_j)''(c) = P_j''(N) > 0,$$
$$(\Delta_N P_j)''(0) = P_j''(N) - P_j''(0) > 0.$$

Hence the relevant tangencies remain exactly double and do not split.

It remains to control the other zeros. For each integer $1 \leq n \leq N$, all possible chord endpoints lie in J, so the starting point lies in the compact interval

$$J_n = \left[-\frac{1}{4}, N + \frac{1}{4} - n\right].$$

By Theorem 6.2, after removing the already treated tangency neighborhoods, every zero of $\Delta_n H$ on J_n is simple and there are only finitely many. Choose disjoint neighborhoods of those zeros on which $(\Delta_n H)'$ has a fixed nonzero sign. The uniform C^1-convergence $P_j \to H$ gives exactly one simple zero of $\Delta_n P_j$ in each such neighborhood. On the remaining compact set, $|\Delta_n H|$ has a positive minimum, so uniform convergence prevents new zeros. Since only finitely many integers $1 \le n \le N$ are involved, one sufficiently large j works simultaneously for all n. Take $P = P_j$. □

Corollary 7.3 (Positive-factor form)**.** *The realizing polynomial in Theorem 7.2 may be chosen in the form*

$$P(x) = x^2(x-N)^2(x-c)^3 R(x),$$

where $R \in \mathbb{Q}[x]$ is strictly positive on $\mathbb{R}$.

Proof. The derivative sign pattern and (7.6) show that the only real zeros of P are $0, c, N$, with multiplicities $2, 3, 2$, respectively. After dividing out those rational factors, the quotient has no real zero and is positive at one, hence every, real point. □

8 Main theorem and transversality

We now combine necessity and realization.

Theorem 8.1 (Extremal classification and rational polynomial realization)**.** *Fix $N \in \mathbb{N}$. A nonnegative integer vector*

$$(a_1, \dots, a_N)$$

occurs as

$$(m_1(F), \dots, m_N(F))$$

for a continuous N-extremal function $F : \mathbb{R} \to \mathbb{R}$ if and only if it is one of the vectors $V_{N,r}$ or $W_{N,r}$ in (5.1)–(5.2). Every such vector is realizable by a polynomial in $\mathbb{Q}[x]$.

Proof. Necessity is Theorem 5.3. For $V_{N,r}$, choose

$$c = N - r - \frac{1}{2}, \qquad e = r + \frac{1}{2}.$$

For $W_{N,r}$, choose

$$c = N - r - 1, \qquad e = r + 1.$$

Corollary 6.3 gives the desired C^2 counts, and Theorem 7.2 preserves all integer multiplicities in a rational polynomial. □

Corollary 8.2 (Transverse classification)**.** *The transverse N-extremal vectors are exactly*

$$V_{N,r}, \qquad 0 \le r < N/2.$$

In particular, there are exactly

$$\left\lceil \frac{N}{2} \right\rceil$$

transverse extremal vectors, and every one has a rational polynomial realization.

Proof. Every $V_{N,r}$ is realized by the construction with noninteger c, e, so all zeros below N are simple. Conversely, after the positive tail orientation, every $\Delta_n F$ has the same sign at both ends. If all of its zeros are simple, their number is even. Therefore an odd coordinate below N is impossible, excluding every W-vector. □

9 What the classification changes

The main theorem replaces three weaker intermediate descriptions.

First, the semigroup theorem records only which lengths may be absent. At extremality, Theorem 4.4 collapses this arithmetic freedom to one integer r: every missing support is a terminal block.

Second, terminal support alone permits formal central allocations

$$(1,3), \quad (2,2), \quad (3,1)$$

in each complementary pair. Theorem 5.3 shows that these choices are not independent. Global compatibility among the difference functions allows at most one odd pair, and it must sit immediately next to the $(4,0)$ terminal band.

Third, a finite list of explicit low-degree examples cannot settle existence for all N. The inverse-branch formula (6.2) solves the realization problem before any polynomial is chosen: it prescribes the full continuum of chord lengths. The polynomialization theorem then transfers the exact integer data while locking the critical-point geometry on the entire real line. Thus no finite numerical verification is needed in the final classification.

10 Further problems

The extremal distinct-chord problem is now closed, but several natural questions remain.

Problem 10.1 (Minimum degree)**.** Determine the least degree of a polynomial realizing each $V_{N,r}$ and $W_{N,r}$.

Problem 10.2 (Uniform degree bound)**.** Does there exist an absolute integer D such that every extremal vector, for every target N, has a polynomial realization of degree at most D? The existence proof in Theorem 7.2 gives finite degree for each parameter pair but no uniform bound.

Problem 10.3 (Realization spaces)**.** For a fixed N and a fixed V- or W-profile, describe the topology and geometry of the realization space modulo translation and positive vertical scaling. In particular, understand how a W-profile sits on the discriminant boundary between transverse V-type regions.

Problem 10.4 (Beyond the minimum)**.** Suppose $m_N > 0$ but $M(F) > 2N - 1$. Determine sharp propagation laws between missing lengths and excess multiplicities, and classify the first few levels above the minimum.

11 Conclusion

For integer horizontal chords, the sharp total $2N - 1$ is only the first layer of the problem. Equality exposes a rigid interaction between the additive structure of missing lengths and the zero structure of the difference functions $\Delta_n F$.

The complete picture is now simple. Missing lengths form one terminal interval. Every central complementary pair is balanced as $(2,2)$, except that one adjacent pair may degenerate to $(3,1)$. These are exactly the V- and W-families, and no other extremal vector is possible. Conversely, every one of these vectors is realized by a rational polynomial.

The main constructive idea is to work in inverse-branch coordinates: horizontal chord multiplicity is then controlled by monotone separation functions rather than by solving a family of high-degree difference equations. The subsequent factorization

$$P' = \Omega U, \qquad U > 0,$$

with two exact moment conditions, preserves the branch geometry while moving into the algebraic category. This combination of global difference constraints, inverse-branch geometry, and positive-weight polynomialization is the structural core of the classification.

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